MMDU MULLANA OFFICIAL SYLLABUS • BPHY-001 • UNIT 2

Unit 2: Electrostatics, Magnetostatics & Electromagnetic Waves (Maxwell's Unified Theory)

Subject: Applied Physics (BPHY-001) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 35 mins

Divergence & curl of electrostatic field, Poisson & Laplace equations, Biot-Savart law, Divergence & curl of static magnetic field, Magnetic vector potential, Equation of continuity, Maxwell's displacement current, Maxwell's 4 equations, Free space wave propagation, Speed of light derivation, Skin depth in conductors, and Poynting theorem.

1. Electrostatic & Magnetostatic Vector Fields & Potentials

1. Electrostatics: Div, Curl & Scalar Potential

- **Gauss's Law (Divergence of E):** Net outward electric flux through any closed surface is proportional to enclosed charge:

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \quad \Leftrightarrow \quad \oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

- **Conservative Nature (Curl of E):** Electrostatic fields are conservative, meaning work done along any closed circuit is zero:

$$\nabla \times \vec{E} = 0 \implies \vec{E} = -\nabla V$$

- **Poisson's and Laplace's Equations:** Substituting $\vec{E} = -\nabla V$ into Gauss's Law:

$$\nabla \cdot (-\nabla V) = \frac{\rho}{\varepsilon_0} \implies \nabla^2 V = -\frac{\rho}{\varepsilon_0} \quad \text{[Poisson's Equation]}$$

In charge-free space ($\rho = 0$), this simplifies to **Laplace's Equation**:

$$\nabla^2 V = 0$$

2. Magnetostatics: Div, Curl & Vector Potential A

- **Gauss's Law for Magnetism:** Magnetic monopoles do not exist; magnetic lines of force form unbroken continuous loops:

$$\nabla \cdot \vec{B} = 0 \quad \Leftrightarrow \quad \oint_S \vec{B} \cdot d\vec{A} = 0$$

- **Ampere's Circuital Law (Static):**

$$\nabla \times \vec{B} = \mu_0 \vec{J} \quad \Leftrightarrow \quad \oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

- **Magnetic Vector Potential A:** Since $\nabla \cdot \vec{B} = 0$, by vector identity the divergence of any curl is zero, allowing us to define $\vec{A}$:

$$\vec{B} = \nabla \times \vec{A}$$

Imposing the **Coulomb Gauge condition** ($\nabla \cdot \vec{A} = 0$), Ampere's Law becomes:

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = -\nabla^2 \vec{A} = \mu_0 \vec{J} \implies \nabla^2 \vec{A} = -\mu_0 \vec{J}$$

2. Displacement Current & Maxwell's 4 Fundamental Equations

1. Inconsistency of Ampere's Law & Continuity Equation

The **Equation of Continuity** expresses conservation of electric charge:

$$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

Taking the divergence of static Ampere's Law ($\nabla \times \vec{B} = \mu_0 \vec{J}$):

$$\nabla \cdot (\nabla \times \vec{B}) = \mu_0 (\nabla \cdot \vec{J})$$

Since divergence of any curl is identically zero ($\nabla \cdot (\nabla \times \vec{B}) \equiv 0$), static Ampere's Law demands $\nabla \cdot \vec{J} = 0$. This violates charge conservation whenever charge density varies with time ($\partial \rho / \partial t \neq 0$, such as during charging of a capacitor!).

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2. Maxwell's Displacement Current Modification

Maxwell eliminated this contradiction by replacing ρ using Gauss's Law ($\rho = \varepsilon_0 \nabla \cdot \vec{E}$):

$$\nabla \cdot \vec{J} + \frac{\partial}{\partial t}(\varepsilon_0 \nabla \cdot \vec{E}) = 0 \implies \nabla \cdot [\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}] = 0$$

Thus, total current density consists of conduction current $\vec{J}$ plus **Displacement Current Density** $\vec{J}_d$:

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

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3. Maxwell's 4 Equations Matrix

Law	Differential Form	Integral Form	Physical Significance
Gauss (E)	$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$	$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$	Charges produce electric divergence.
Gauss (B)	$\nabla \cdot \vec{B} = 0$	$\oint \vec{B} \cdot d\vec{A} = 0$	Non-existence of isolated magnetic monopoles.
Faraday	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$	Changing magnetic field induces circular electric field.

| **Ampere-Maxwell** | $\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$ | $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d}{dt} \int \vec{E} \cdot d\vec{A}$ | Conduction + displacement currents generate magnetic fields. |

3. Wave Propagation in Free Space & Speed of Light Derivation

1. Derivation of Free-Space Wave Equation

In source-free vacuum ($\rho = 0, \vec{J} = 0$):

1. $\nabla \cdot \vec{E} = 0$

2. $\nabla \cdot \vec{B} = 0$

3. $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

4. $\nabla \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$

Taking curl of Faraday's Law (3):

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B})$$

Applying vector identity $\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$, and substituting (1) and (4):

$$0 - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}(\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}) \implies \nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

Similarly for magnetic field: $\nabla^2 \vec{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$.

Comparing with 3D classical wave equation $\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$ establishes that electromagnetic radiation travels with speed:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}} \approx 2.998 \times 10^8 \text{ m/s}$$

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2. Transverse Nature & Intrinsic Impedance

- Wavevector $\vec{k}$ is perpendicular to both fields: $\vec{k} \cdot \vec{E} = 0$ and $\vec{k} \cdot \vec{B} = 0$.
- Amplitude ratio: $E_0/B_0 = c$.
- **Characteristic Free-Space Impedance (Z_0):**

$$Z_0 = \frac{E_0}{H_0} = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12}}} \approx 376.73\ \Omega \approx 377\ \Omega$$

4. Propagation in Conducting Media, Skin Depth & Poynting Theorem

1. Attenuation & Skin Depth Derivation in Conductors

In a conducting medium with conductivity σ, permittivity ε, and permeability μ, conduction current $\vec{J} = \sigma\vec{E}$. The wave equation becomes:

$$\nabla^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

For a harmonic plane wave $E(z,t) = E_0 e^{i(\omega t - \gamma z)}$, the complex propagation constant is $\gamma = \alpha + i\beta$. For a **good conductor** ($\sigma \gg \omega\varepsilon$):

$$\gamma = \sqrt{-i\omega\mu\sigma} = \sqrt{\omega\mu\sigma}\, e^{-i\pi/4} = \sqrt{\frac{\omega\mu\sigma}{2}} - i\sqrt{\frac{\omega\mu\sigma}{2}}$$

Hence, the spatial attenuation constant α is:

$$\alpha = \sqrt{\frac{\omega\mu\sigma}{2}} = \sqrt{\pi f\mu\sigma}$$

The electric field attenuates exponentially: $E(z) = E_0 e^{-\alpha z} = E_0 e^{-z/\delta}$.

✅ **FUNDAMENTAL LAW / RULE:**
Rigorous Definition: Skin Depth (δ)

Skin depth is the penetration depth into a conducting material at which the electromagnetic wave amplitude drops to $1/e \approx 36.8\%$ of its surface value:

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega\mu\sigma}} = \frac{1}{\sqrt{\pi f\mu\sigma}}$$

Engineering Impact: At radio frequencies (RF), current is confined strictly to a thin microscopic skin, requiring silver-plated conductors and explaining underwater RF blackout.

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2. Poynting Vector & Work-Energy Theorem

From Maxwell's curl equations, taking the scalar product:

$$-\nabla \cdot (\vec{E} \times \vec{H}) = \frac{\partial}{\partial t}\left[\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2\right] + \vec{J} \cdot \vec{E}$$

Integrating over closed volume V bounded by surface S:

$$-\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{A} = \frac{\partial}{\partial t}\int_V u_{\mathrm{EM}} dV + \int_V (\vec{J} \cdot \vec{E}) dV$$

- **Poynting Vector:** $\vec{S} = \vec{E} \times \vec{H}$ represents instantaneous power flow per unit area ($\mathrm{W/m^2}$).
- **Physical Law:** Net rate of electromagnetic energy flowing inwards across surface S equals time rate of increase of stored field energy plus Ohmic Joule dissipation rate.